

DESIGN DOCUMENT: Pose Classification & Object Detection System

Project Developer: AI Engineering Team

Model Architecture: YOLOv8n (Nano)

Dataset: Ogmen Robotics Dog Dataset (180 Images: 150 Train / 30 Test)

1. Executive Summary

This design document details the engineering decisions, architectural choices, and validation results of a custom-trained object detection and pose classification system. The objective of this system is to accurately localize and classify dog postures into three distinct target classes: **standing**, **sitting**, and **lying**. Utilizing a transfer-learning approach with **Ultralytics YOLOv8n**, we established a lightweight, real-time edge-deployable pipeline achieving **80.00% overall classification accuracy** on a dedicated 30-image test set.

2. Architecture Choice & Engineering Decisions

Why YOLOv8n (YOLOv8 Nano)?

- **Real-time Edge Efficiency:** Robotic platforms (such as those engineered by Ogmen Robotics) demand high-throughput, low-latency processing. YOLOv8n consists of only **3.0 million parameters** and requires **8.1 GFLOPs**, making it ideal for CPU/GPU edge deployment, ensuring inference latencies well below **15ms per frame**.
- **Small dataset:** the dataset contained only 150 images, models like yolov8 and yolov8m will mostly overfit.
- **Single-Stage Unified Pipeline:** Unlike traditional two-stage systems (which first detect a dog and then pass the cropped region to a secondary CNN classifier), YOLOv8 predicts the bounding box coordinates and the posture classification simultaneously in a single forward pass. This prevents compounding latency overhead.
- **Transfer Learning Robustness:** By leveraging pretrained COCO weights, the model's backbone starts with a deep representational understanding of spatial features, textures, and edges, allowing it to converge rapidly on a relatively small dataset (150 training samples).

Data Quality Control & Engineering Interventions

During system initialization, two critical data-engineering flaws were identified and corrected:

1. **Label Index Correction:** The original label files contained indexing bugs where multiple classes mapped to the same index (0). We built a systematic re-mapping pipeline to correctly index labels matching our physical pose schema: 0: standing, 1: sitting, and 2: lying.
2. **Coordinate Normalization:** Training label coordinates from different splits were misaligned (some containing unnormalized absolute pixel bounds). A coordinate-normalization script was integrated to scale all bounding boxes strictly between 0.0 and 1.0 based on image shapes prior to training, preventing gradient explosions.

3. Test Results & Performance Metrics

Overall Performance

- Overall Accuracy: 80.00%** (24 out of 30 test images classified perfectly).
- Mean Average Precision (mAP@50): 85.3%** across all classes.
- Speed Metrics:** Preprocess: **0.2ms**, Inference: **2.2ms**, Postprocess: **1.1ms** per image.

Per-Class Metrics

Class	Ground Truth Count	Correct Predictions	Precision	Recall	F1-Score
Standing	10	8	80.00%	80.00%	0.80
Sitting	10	9	75.00%	90.00%	0.82
Lying	10	7	87.50%	70.00%	0.78
No Detection	0	0	100.00%	0.00%	0.00

Note: Precision/Recall ratios derived strictly from confusion-matrix matching of localization predictions.

4. Error Analysis & Edge Case Failures

While the model performs exceptionally well, a granular audit of the 6 misclassified frames revealed specific conditions under which the model failed:

1. Ambiguous Postural Boundaries (Sitting vs. Standing / Sitting vs. Lying):

- Failure Case:* standing_test_0007.jpg and standing_test_0010.jpg were classified as sitting.
- Root Cause:* In these images, the dog exhibits a highly slouched "semi-standing" or transitioning posture. Due to the vertical compression of the torso, the bounding-box aspect ratio closely resembles that of a typical sitting pose.
- Failure Case:* sitting_test_0002.jpg was classified as lying.
- Root Cause:* The dog was sitting but heavily leaning back on a soft surface (couch), mimicking the low-height aspect ratio of lying profiles.

2. Severe Occlusion & Environmental Camouflage:

- Failure Case:* lying_test_0002.jpg resulted in no_detection.
 - Root Cause:* The subject was lying on a dark patterned rug with severe occlusion (only the dog's head and neck were clearly visible).
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